

[DC] Impact of Mixed Reality on Spatial Perception and Affordances

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ABSTRACT

This study examines the impact of mixed reality environments on spatial behavior. In a first study, we investigated a navigation task, specifically a goal-directed task. Participants performed this task under both real-world and mixed reality conditions, navigating through doors of varying widths. The main objective was to evaluate how the technological limitations of mixed reality headsets (including restricted field of view, resolution, and refresh rate) influence navigation behaviors. Two hypotheses were tested: (1) mixed reality modifies behavioral adjustments compared to a real-world environment, and (2) distance perception in mixed reality is underestimated, leading to compensatory behavioral adjustments, particularly in deceleration and shoulder rotation strategies. The key variables analyzed included deceleration threshold (the distance at which participants reduce their speed before passing through the door) and shoulder rotation threshold (the distance at which they begin adopting a lateral gait). The results aim to shed light on the effects of immersive environments on spatial perception and motor adaptation, with implications for the design of navigation interfaces in mixed reality systems.

Index Terms: spatial perception, affordances, mixed reality, pass-through.

1 CONTEXT

In the industrial maintenance sector, Airbus Helicopters is exploring the use of mixed reality (MR) to address the challenges of accessibility and feasibility of maintenance tasks. By integrating virtual environments with real-world settings, this technology offers the possibility of simulating complex scenarios while reducing dependence on physical components. However, the effectiveness of these environments relies on operators' ability to maintain accurate spatial perception and awareness, defined as the ability to perceive and locate body position and movements in a mixed space. This project aims to assess the ecological validity of mixed environments, by analyzing their influence on human behavior, with a view to improving interactions in these environments to make them more consistent with behaviors observed in real-life conditions.

2 INTRODUCTION

The ecological approach to visual perception, also known as the theory of direct perception, emphasizes the importance of optical flow in providing information about the environment [3]. Linked to movement, this perception uses the optical flow generated by the agent or the environment to guide action [7]. This dynamic is closely linked to the perception of affordances, a concept introduced by [1], which refers to action opportunities arising from the interaction between environmental properties and an individual's physical abilities. The perception of affordances depends on the morphology and physical skills of the user and includes two

types of information: body-scaling, which compares environmental dimensions to the individual's physical structure, and action-scaling, which assesses the dynamic capabilities needed to complete a task [1]. This approach highlights the physicality of perception, emphasizing the role of dynamic information in adjusting motor actions in space [7].

The perception of affordances relies on the integration of multiple types of visual information, distinguished by their functions in evaluating space and possible actions. Processes such as parallax and optical flow provide information about spatial structure and motion in the environment [10]. Others, like tau, focus on the temporal aspects of dynamic interactions. Parallax occurs when nearby objects appear to move faster than distant ones in response to the observer's movements, enhancing depth and relative distance perception. Meanwhile, optical flow, generated by the individual's or environment's movement, provides essential cues about direction, speed, and spatial geometry, facilitating motor adaptation [5]. However, spatial information alone is insufficient to guide action. Tau, as a temporal cue, addresses this need by anticipating the critical moment when an object reaches a given position. This anticipation is crucial for synchronizing motor adjustments in tasks like navigation or obstacle avoidance. By combining these elements, parallax and optical flow offer a spatial representation of the environment, while tau manages the temporal aspects of interactions. This connection between spatial and temporal information is vital, particularly in immersive environments where spatial complexity requires rapid and precise motor adaptations.

In a MR context, where virtual elements are interactively integrated with the real-world to allow dynamic interaction between these two realms, the principles of the ecological approach find direct application. This technology enables users to engage with physical objects while simultaneously interacting with digital objects superimposed on the environment [8]. In this context, the technical specifications of mixed-reality headsets can significantly impact the perception of visual parameters such as parallax, tau, and optical flow. Elements like resolution and refresh rate directly influence the clarity and temporal coherence of perceptual cues [6]. Limited resolution or low refresh rates can impair the perception of fine details and introduce temporal delays, disrupting the assessment of depth and motion dynamics. Additionally, a restricted field of view reduces the richness of available visual information, affecting the perception of the overall environment and affordances [2].

These factors influence distance perception in mixed environments, with a general tendency to underestimate distances. This underestimation poses a significant challenge in tasks requiring accurate distance evaluation, such as navigation or interaction tasks [4]. Perceived affordances, increasingly used as an objective measure of the perception of virtual spaces, generally show good perceptual fidelity between real and virtual objects while identifying certain differences. Action-based measures, such as individuals adjusting to pass through openings based on shoulder width [4], are a reliable approach for evaluating this fidelity. However, studies like [9] indicate that crossing judgments in MR are often underestimated due to technical limitations such as restricted field of view [2].

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3 RESEARCH QUESTIONS AND HYPOTHESES

This study explores the impact of MR on spatial adjustment behaviors during a navigation task. Two research questions are addressed. First, to what extent do the technological limitations of MR headsets influence behavioral adjustments in MR environments compared to real-world environments? This question is based on the hypothesis that the restricted field of view (FOV) and the visual rendering of the headset's cameras disrupt environmental perception, thereby altering navigation behaviors. Second, the study aims to determine whether virtual affordances in MR environments lead to distance underestimation, prompting anticipatory behavioral adaptations compared to a real-world environment. This is based on the hypothesis that MR environments distort distance perception, resulting in compensatory adjustments in user actions.

4 METHOD

This study consists of two phases, each involving a navigation task where participants pass through a wooden structure with two doors representing an opening. Starting at point A, they pass through the opening (30–100 cm in 10 cm increments) to reach point B. Each opening size is tested five times (8 opening sizes $\times$ 5 repetitions) to evaluate behavioral adjustments based on the opening size and experimental conditions.

In the first phase of this study, the task is performed under two conditions: (1) wearing a MR headset (Meta Quest 3, Meta Platform, Inc., United States) to visualize the real-world environment without virtual objects and (2) without wearing the headset. This phase investigates the impact of camera-based visual rendering and reduced FOV on affordance perception and navigation behavior.

In the second phase, the task is repeated under two additional conditions: (3) a real wooden structure with superimposed virtual doors and (4) a fully virtual structure and doors. The virtual components were created using Blender and Unity. This phase explores how increasing the amount of virtual information in the headset affects distance perception and navigation behavior.

The positions of participants' heads, shoulders, and feet were recorded throughout the task to analyze their motor strategies. Position data were captured using Active Traqr sensors (Qualisys, AB, Sweden) along three axes: front–back (X-axis), left–right (Z-axis), and up–down (Y-axis).

5 RESULTS AND EXPECTATION

This study aims to examine participants' adaptive behavior under different experimental conditions, focusing on variables related to walking speed and shoulder rotation during the task. The first variable of interest is the deceleration threshold, which measures the distance at which participants begin to slow down as they approach the doorways. The second variable is the shoulder rotation threshold, which identifies the distance at which participants start adopting a sideways walking strategy by rotating their shoulders to navigate through the door opening. The analysis emphasizes participants' behavioral regulation across different conditions. Additional variables to be analyzed include the distance between participants' shoulders and the door frames as they pass through. Other variables include the passage time and the number of contacts detected with the doors. The main goal of this study is to determine whether the affordances offered by a MR environment are comparable to those of a real-world environment. Specifically, the study seeks to explore potential differences between these two types of environments and to quantify these differences. The first stage of this study aims to understand the extent to which MR environments can effectively substitute real-world environments. This study will then enable us to try and reduce this gap by working on improving MR interactions through the use of sensory feedback. This feedback, whether visual, auditory or vibratory, could be integrated to optimize perception and interaction with virtual

objects. The ultimate goal is to use the detection of contact with virtual objects via these sensory feedbacks to improve perception and encourage more natural and effective interaction with MR environment.

As part of the Doctoral Consortium, two areas will be discussed:

1. Impact of headset technical specifications: The limited FOV can alter the perception of visual cues such as parallax. Investigating the effects of further reducing the FOV could provide insights into optimizing immersive experiences.
2. Methodological limitations of the experimental phase: In this first study, the participants were not informed about the existence of contacts when passing through the doors, leaving uncertainty about their perception of obstacle clearance. The next stage of this research will involve integrating sensory feedback to signal these contacts, thereby reinforcing the perception of interactions and improving behavior in MR, making it closer to that observed in real-life conditions.

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